

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 70%. We keep your highest score.

Next item →

1.

Which of the following expressions is equivalent to $P \vee (Q \wedge R)$

1 / 1 point
- ☐

$(P \vee Q) \wedge (P \wedge R)$
- ☐

$(P \wedge Q) \vee (P \wedge R)$
- ☒

$(P \vee Q) \wedge (P \vee R)$
- ☐

$P \vee Q \wedge R$
-  Nice work
2.

Which one is the negation of:

1 / 1 point
- $(P \wedge \neg Q) \rightarrow P$
- ☐ True
- ☐ Q
- ☐ $\neg Q$
- ☒ False
-  Nice work
3.

Which of the following statements is a tautology?

1 / 1 point
- ☒ $P \vee \neg P$
- ☐ $P \wedge \neg P$
- ☐ $\neg(P \vee Q)$
- ☐ $P \rightarrow Q$
-  Nice work
4.

Determine which of the following is a contradiction (an expression is a contradiction if it is always False)

1 / 1 point
- ☐ $\neg(P \wedge \neg P)$
- ☒ $(P \wedge \neg Q) \wedge (\neg P \vee Q)$
- ☐ $P \wedge \neg Q \wedge (Q \vee P)$
- ☐ $P \vee \neg Q$
-  Nice work
5.

What is the negation of $P \wedge (Q \vee R)$?

1 / 1 point
- ☒ $\neg P \vee (\neg Q \wedge \neg R)$
- ☐ $\neg P \vee \neg Q \vee \neg R$
- ☐ $P \wedge \neg Q \wedge \neg R$
- ☐ $\neg P \wedge (\neg Q \vee \neg R)$
-  Nice work
6.

What is the truth value for p, q if we want $(p \rightarrow \neg q) \vee (\neg p \wedge q)$ to be False?

1 / 1 point
- ☐ $p = F, q = T$
- ☐ $p = T, q = F$
- ☒ $p = T, q = T$
- ☐ $p = F, q = F$
-  Nice work
7.

Which of the following is equivalent to the expression $\neg(\neg P \wedge Q)$

1 / 1 point
- ☐ $\neg P \vee \neg Q$
- ☐ $\neg P \wedge Q$
- ☐ $\neg P \vee Q$
- ☒ $P \vee \neg Q$
-  Nice work
8.

Assume $q \leftrightarrow \neg p$ is False. Which of the following choices is always True:

1 / 1 point
- ☐ $p \wedge q$
- ☐ $p \rightarrow \neg q$
- ☐ $p \vee q$
- ☒ $p \rightarrow q$
-  Nice work